

Responsible AI

Privacy, consent, safety and responsible data use

Sauti Yetu | Multilingual voice triage support for primary healthcare

First principle. Sauti Yetu is decision support. It produces hints to aid a clinician's judgement. It does not diagnose and it does not make a care decision on its own.

Why this note exists

Sauti Yetu handles a patient's health complaint in the patient's own voice and language. That creates responsibilities at every boundary: what the system collects, which component receives it, how long it remains available, how a suggestion can be challenged, and how a person is treated when the system is uncertain.

This note translates those responsibilities into implementation commitments for clinicians, reviewers and the people whose voices the system hears. Each commitment is paired with the mechanism described for the current prototype and with the evidence still needed before a real clinic deployment.

Audience	What the note should make clear
Patients	What is recorded, what is shown back, how a response is communicated and how withdrawal should work in a real clinic.
Clinicians	Which outputs are suggestions, where the evidence comes from and where professional judgement remains required.
Reviewers and builders	What the codebase claims to enforce, what must be tested and which limitations remain open.

Status and scope

Working prototype, hackathon scope. The commitments below describe the supplied Responsible AI note and its stated codebase mechanisms. The code, clinical review and local health-data requirements remain the source of truth if this note ever drifts from implementation.

Decision support only. This document does not establish clinical validation, regulatory clearance or suitability for unsupervised care.

A responsible system is a set of boundaries

The commitments work together. Data minimisation limits exposure; consent and transparency give people visibility; safety and human oversight constrain decisions; provenance makes outputs inspectable; bounded generation limits what an external model can do; fairness keeps the system aligned with how people communicate.

Commitment	Prototype mechanism	Review question
Data minimisation	Transient audio, structured context downstream, optional patient context and minimum external payloads.	Does each component receive only the fields needed for its task?
Privacy and retention	Process-local in-memory sessions, explicit deletion and local-first models.	Where else could data remain: logs, caches, clients or providers?
Consent and transparency	Transcript shown back; replies displayed and spoken; deletion endpoint available.	Can a patient understand, correct or withdraw the processing?
Safety	Escalate-only urgency, red flags, human fallback and fail-fast knowledge-base validation.	What happens when evidence is missing, wrong or contradictory?
Explainability	Source-tagged rules, reasoning graph and deterministic decision path.	Can a clinician trace each hint to evidence and challenge it?
Bounded generative AI	GPT-4o confined to documentation with a strict prompt and context boundary.	Can generated text introduce facts or change the clinical result?
Fairness and access	Code-switching support, local models, patient-language replies and WER/CER reporting.	Which speakers, settings and language mixes remain underrepresented?

How to read the safeguards

A design mechanism is a control point, not proof of performance. For example, a local model supports a data-stays-here deployment only when the model weights, runtime, network boundary and failure behavior have been tested. A citation identifies the rule source; it does not make an individual output correct.

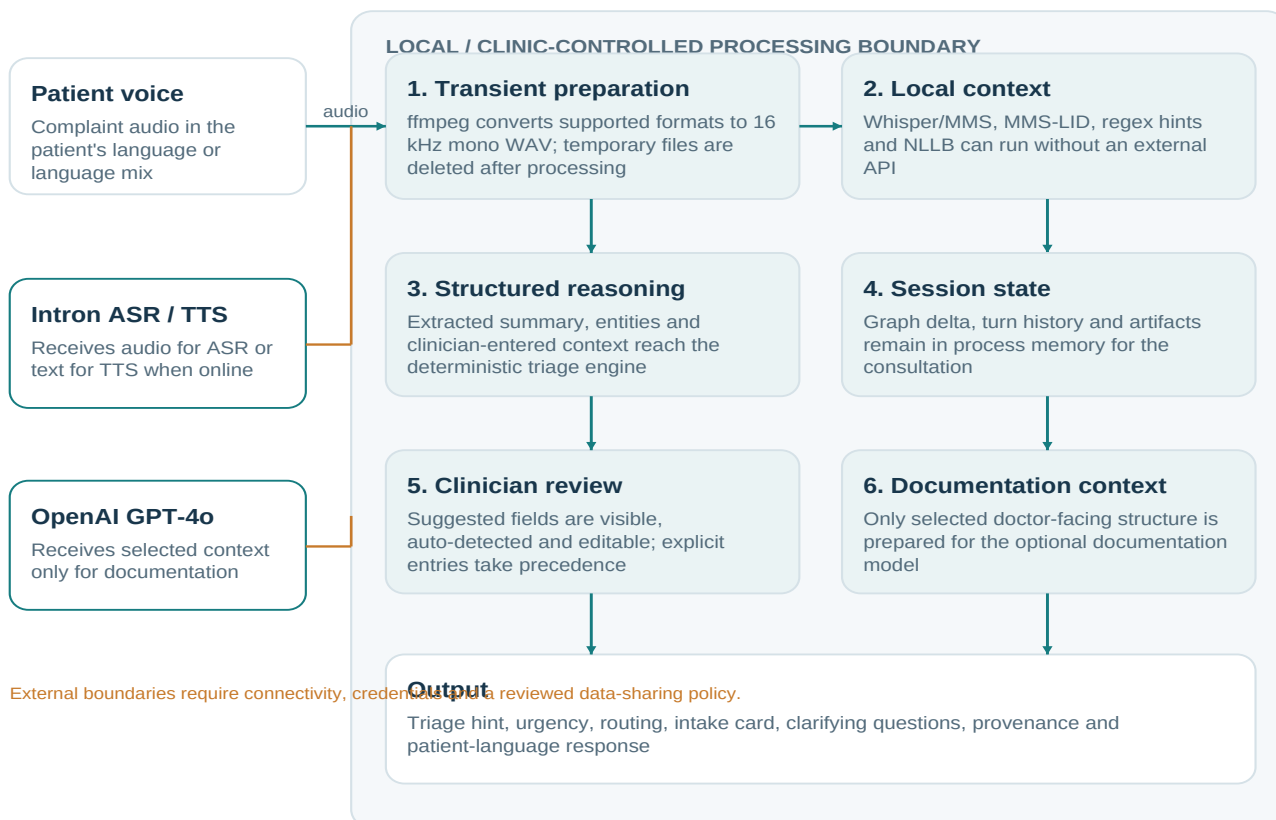
Accountability boundary. The system can make uncertainty visible and constrain its own behavior. A qualified health worker remains responsible for interpreting the output, checking the patient and deciding the next clinical action.

The central operating pattern

Collect the minimum -> preserve uncertainty -> apply reviewable rules -> show the evidence -> keep a human in control -> delete the session when it ends.

Move only what each layer needs

The architecture separates raw speech from clinical structure. Audio is used for perception, context is used for extraction and reasoning, and selected doctor-facing structure is used for optional documentation. This separation reduces unnecessary propagation of sensitive data and makes each boundary explicit.



The diagram describes intended data minimisation; logs, OS caches and provider retention require separate controls.

Data categories and controls

Data	Primary use	Control and qualification
Raw audio	Speech recognition and, when requested, speech synthesis input.	Converted for processing and described as transient. Cleanup on errors and interruption paths needs implementation verification.
Transcript and extractions	Context extraction, deterministic triage and clinician review.	Raw audio is excluded from the reasoning layer; speech recognition errors can still affect downstream outputs.
Age, sex, pregnancy and vitals	Gate specific rules or evaluate entered readings.	Optional patient context is additive. Vitals are evaluated only when entered by a clinician and are never inferred from audio.
Session graph and artifacts	Maintain consultation history and support documentation.	Stored in process memory, with explicit DELETE. Logs, client state and provider handling require separate review.
External payloads	ASR/TTS and optional documentation.	Intron receives audio for ASR or text for TTS; GPT-4o receives selected context only. Connectivity and provider policies apply.

Why minimisation improves safety

Limiting data flow reduces the number of places where a mistake can spread. It also makes review easier: a reviewer can ask whether an external request contains audio, identity information, clinical structure or only the text needed for the requested feature. Minimisation should be checked at request construction, logging and error handling, not inferred from the interface alone.

03 / PRIVACY AND CONSENT

Retention is a lifecycle, not a promise

The current design treats a consultation as an ephemeral session. That supports a short retention window, but it still requires an explicit lifecycle: start, use, review, end and verify deletion. In-memory storage limits persistence inside the application process; it does not automatically govern operating-system files, logs, browser state or external providers.

Session lifecycle

Stage	Expected behavior	Control to verify
Start	Create one session for one consultation.	Session identifier is generated without exposing identity data in the identifier or response.
Process	Append turn history, graph deltas and artifacts in process-local memory.	No database write; access is limited to the intended consultation and backend process.
Review	Show transcript, suggestions, evidence and translated responses to the clinician and patient as appropriate.	Corrections and explicit clinician values are visible and take precedence.
End	Call DELETE /api/session/{id} to drop the stored consultation.	Deletion returns a clear status and removes the session from the active store.
Restart or failure	A process restart clears the in-memory store; temporary files should be cleaned.	Test cleanup on success, timeout, cancellation, crash and worker restart paths.

Consent and transparency in the patient interaction

The patient should be told that their complaint is being recorded and processed before the recording begins. The prototype supports transparency by surfacing the transcript, displaying the worker reply and making the reply available as audio in the patient's language. A real clinic still needs a consent workflow that records the choice, explains the purpose and provides a path to withdraw.

Withdrawal is an operational action. The product should stop processing, delete the active session and communicate what has happened. The prototype's explicit deletion endpoint is a building block; clinic policy must define how this interacts with logs, backups and external-service retention.

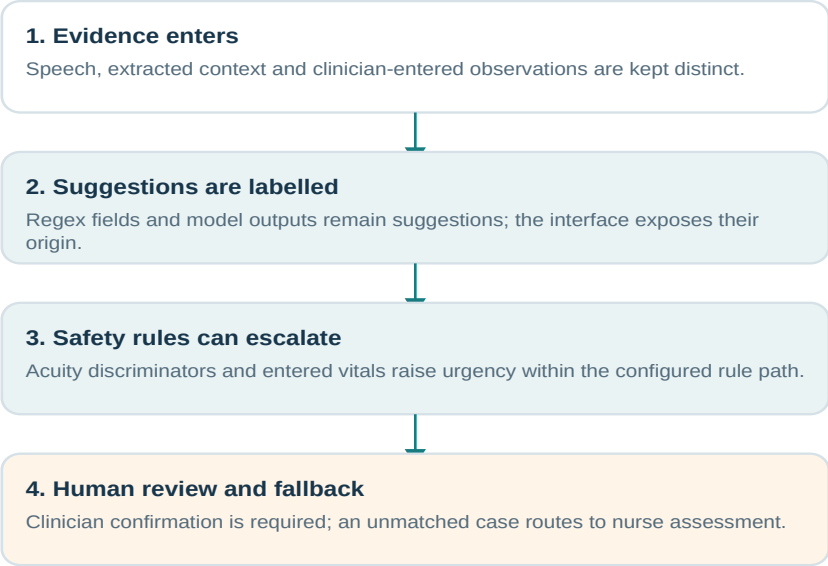
Secrets and outputs

API keys are described as living in a git-ignored .env file and excluded from transcripts, logs, reports and benchmark artifacts. This should be verified with secret scanning and failure-path logging tests. A key being absent from a response does not by itself prove that a sensitive transcript was handled safely.

04 / SAFETY

Constrain the decision path and expose uncertainty

Safety is implemented through several controls rather than one model score. The system separates suggestions from clinician input, uses a reviewable knowledge base, escalates urgency within the configured rule path and sends unmatched cases to human assessment.



SAFETY PROPERTY The pipeline reduces silent under-triage risk, but cannot recover evidence lost in upstream transcription or extraction.

Risk	Control described in the brief	Residual question
Under-triage	Acuity discriminators and entered vital checks can escalate urgency and preserve red flags.	Was the relevant symptom recognized by upstream ASR or extraction?
Unsupported complaint	Fallback routes to "Undetermined - needs nurse assessment" rather than guessing.	Does the UI make the handoff prominent and easy to act on?

Risk	Control described in the brief	Residual question
Incorrect auto-filled context	Regex extraction is suggest-only; explicit clinician input wins; auto_detected fields are flagged.	Can a clinician see the original evidence and correct the field before proceeding?
Knowledge-base error	Knowledge base is validated on boot; malformed data stops the app.	Are clinical mappings reviewed, versioned and monitored after release?
Service failure	ASR transport errors are normalized into clear messages.	Does the client provide a safe retry or human-assessment path for every failure?

What escalate-only means

The rule path is designed so configured discriminators and measured vitals can raise urgency rather than lower it. This is a useful safety property within that path. It cannot guarantee that all urgent evidence was captured, nor can it replace a clinician who notices a concern that the rules do not cover.

Human-in-the-loop is visible in the product

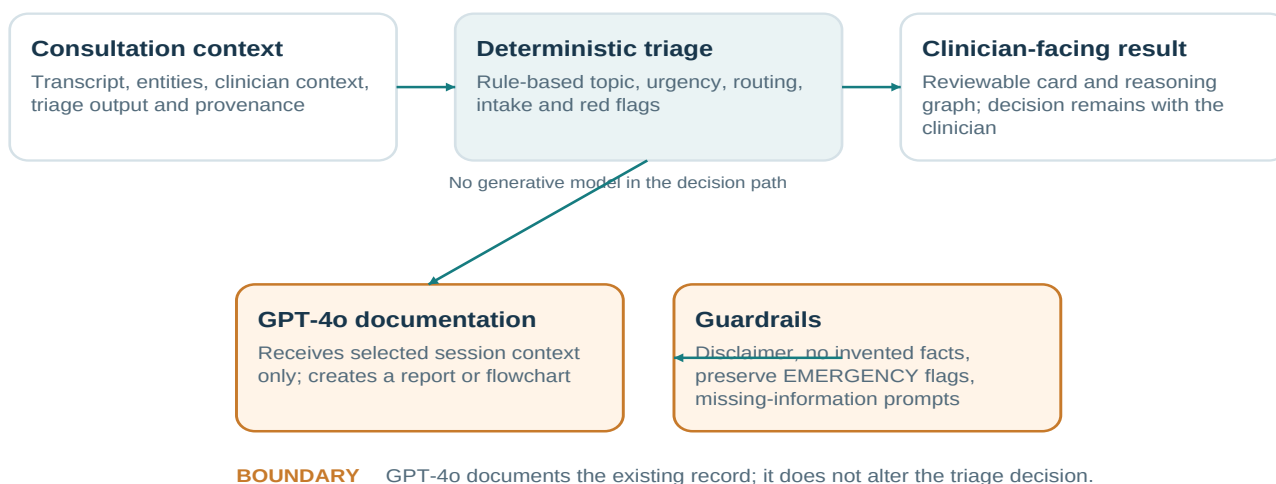
The clinical card, suggested fields, provenance badge and reasoning graph should make review an active step. The interface should distinguish three states: system suggestion, clinician-confirmed fact and unresolved information. That distinction supports safer handoffs and clearer audit trails.

05 / EXPLAINABILITY

Make every clinical hint inspectable

Explainability in Sauti Yetu has two layers. The knowledge base explains why a rule exists; the reasoning graph explains how the current consultation matched that rule. Together they let a clinician move from a patient statement to a finding, topic, urgency or red flag and then to the cited source.

Provenance field	Role in the review
evidence_level	Signals whether the assertion is provisional, expert-derived, guideline-based or dataset-based.
review_status	Shows whether the mapping remains unreviewed or has received clinician review.
mapped_scale	Identifies the clinical scale or framework used for the mapping, such as WHO IITT or SATS.
ref	Provides a human-readable reference that a reviewer can understand without inspecting code.
url	Links the assertion to its source for traceability, with licensing and version checks still required.



Bounded generative AI

The documentation agent is separated from the clinical decision path. It receives selected session context and turns an existing consultation into a report or flowchart. The described prompt requires a decision-support disclaimer, forbids invented symptoms, vitals or conditions, preserves EMERGENCY flags, asks for missing information and writes in the clinician's language.

Boundary	Expected behavior
Input boundary	No raw audio; selected doctor-facing session context only.
Task boundary	Documentation and visualization; no triage decision or diagnosis.
Content boundary	Use recorded facts, name missing information and preserve emergency markers.
Review boundary	Clinician checks the artifact against the session before using it.

Prompt constraints reduce predictable failure modes, but generated text remains generated text. The artifact should be treated as a draft record whose claims are checked against the transcript, intake card and reasoning graph.

06 / FAIRNESS AND ACCESS

Design for how people actually communicate

The system is built around code-switching between Swahili and English rather than requiring a single polished language. Patients can speak in their own language; clinicians can read in theirs; the product can provide a translated response when the selected language pair is supported.

Design choice	Benefit	What still needs evidence
Code-switching first	Preserves meaning in ordinary regional speech patterns and mixed-language complaints.	Coverage across accents, ages, dialects, speaking rates and real clinic conditions.
Patient-language response	Reduces the communication burden on patients during a stressful encounter.	Translation quality, supported language pairs and behavior for passthrough or unknown pairs.

Design choice	Benefit	What still needs evidence
Local, key-free models	Supports clinics with intermittent connectivity and reduces dependence on paid APIs for the core loop.	Verified network-isolated performance, hardware requirements and failure behavior.
WER and CER together	CER can reveal spelling and word-boundary effects in agglutinative languages such as Swahili and Kinyarwanda.	Independent references, normalization rules and clinical meaning checks.
Honest benchmark reporting	Makes sample size, reference provenance and subset limits visible.	Broader speakers, settings and paired clean/noisy data before deployment claims.

Dignity is a product behavior

A safe fallback should preserve the patient's dignity: explain that more assessment is needed, avoid presenting an uncertain guess as a diagnosis and give the health worker a clear next action. Language support should increase participation in care, not hide the system's uncertainty behind fluent text.

Fairness evaluation plan

Track transcription and downstream errors by language mix, accent, age group, gender where ethically and lawfully collected, device, environment and clinical category. Compare missed urgency and missing entities, not only average WER. Investigate whether a model appears strong because a subset shares its recording conditions or reference conventions.

Access principle. Offline-capable design is an access strategy, not a completed guarantee. It becomes meaningful only after a local-only run shows which features continue to work, how long they take and how the system behaves when a service is unavailable.

Move from prototype commitments to deployment evidence

The Responsible AI note is strongest when each commitment has an owner, a test and a visible failure response. The following gate turns the note into an implementation checklist for the next stage.

Area	Before a clinic pilot	Evidence to retain
Consent	Add an informed-consent and withdrawal workflow in the client and clinic process.	Consent wording, displayed notice, withdrawal action and deletion confirmation.
Clinical knowledge	Obtain clinician sign-off on topics, urgency mappings, thresholds and red flags.	Versioned knowledge-base review record, reviewer identity and change history.
Privacy	Define secure hosting, access control, transport protection, logging and provider retention settings.	Threat model, configuration review, log sampling and provider terms.
Offline mode	Run with network access disabled and preinstalled local models; record warm and cold behavior.	Command, hardware, model hashes, successful and failed clips, latency and feature availability.
Noise and fairness	Evaluate matched clean/noisy clips and diverse speakers across supported language mixes.	Audio hashes, SNR or noise metadata, subgroup results, error analysis and remediation decisions.
Human oversight	Train staff to read suggestions, verify fields and handle fallback or service failure.	Workflow guide, escalation route, audit sample and incident process.
Documentation	Review generated reports before they enter a record or inform care.	Source session, artifact version, reviewer confirmation and correction history.

Known limitations

This is a prototype with a small and partly unreviewed knowledge base. Benchmark results indicate trends rather than clinical validity. Clinical guidance adapted from third-party sources requires attribution, licensing review and local clinical confirmation before paid deployment. The current architecture also describes an in-memory process-local store, which needs a deliberate strategy if multiple workers or longer-lived services are introduced.

If implementation and note diverge

Source of truth. If a claim in this note ever drifts from the code or from clinical safety requirements, treat the implementation and the safety review as the source of truth, correct the system, and update this document.

Report-ready statement

Sauti Yetu is designed as a clinician-controlled, multilingual decision-support system. Its responsible AI approach combines data minimisation, ephemeral session handling, transparent patient interaction, conservative rule-based triage, cited provenance, bounded documentation generation and support for code-switched communication. These mechanisms make the prototype reviewable and safer to extend; they do not replace clinical validation, deployment controls or professional judgement.

Source: supplied Responsible AI note, sections 1-8. Module names and mechanisms are reported from the note. Implementation qualifications and deployment checks are editorial analysis for this brief. No repository inspection or live-system testing was performed.

Minimum evidence to retain

A responsible AI claim is easier to review when the mechanism, test evidence, owner and known limitation are recorded together. Use this page as a compact handoff between the builders, clinical reviewers and deployment team.

Evidence area	Retain before a pilot or public claim
Privacy	Data-flow map, request samples, log review, temporary-file cleanup test and provider retention settings.
Safety	Clinician sign-off on the knowledge base, red-flag test cases, fallback checks and incident response route.
Offline	Network-disabled command, model and dependency hashes, hardware, cold and warm timing, failures and unavailable features.
Fairness	Paired subgroup results by language mix, accent and environment, with error analysis and remediation decisions.
Transparency	Consent notice, transcript correction path, translated reply display and session deletion confirmation.

Release gate. A responsible AI claim is ready to share when the mechanism, test evidence, owner and known limitation are recorded together. If any one is missing, describe the capability as a design intention and keep the review open.

Pilot sign-off record

Control area	Owner	Status / date	Evidence reference
Consent and withdrawal	To be assigned	Open	Notice, workflow and deletion test
Clinical knowledge review	To be assigned	Open	Versioned clinician sign-off
Privacy and provider handling	To be assigned	Open	Threat model and configuration review
Offline and local-only run	To be assigned	Open	Command, hashes, hardware and results
Fairness and noise evaluation	To be assigned	Open	Matched clips and subgroup analysis
Human oversight and incidents	To be assigned	Open	Training, escalation and audit record

The status values are placeholders for the next review. A completed record should point to the exact test output, review note or configuration artifact that supports each claim.